The New™ and Improved™ Guide to Timeline

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Disclaimer: ESL, sorry in advance
This guide is about how I do things. Different people have different experiences and do things differently.
This guide will assume the reader has a basic understanding of the chara studio.
Some sections will contain r-18 content.

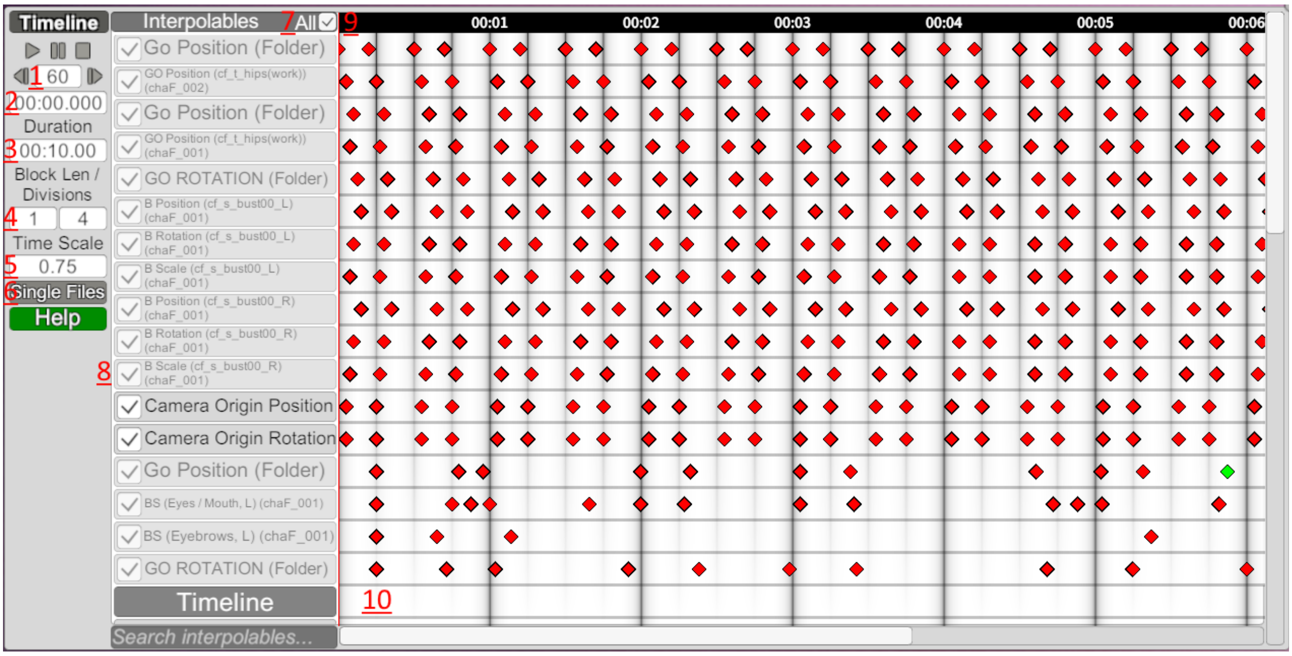
This guide for Timeline is something I have been meaning to make for quite some time. I have picked up animating with this plugin about 18 months ago and have tried my hands at well over 100 scenes. I have thus acquired significant experience animating with timeline and have developed my own workflow to make good quality scenes quickly. This document aims at giving you what I have learned and get you up to speed on timeline so you may attempt your own projects.

This guide will be divided into three sections. First, a quick overview of Timeline, how it works as well as some useful features not everyone knows. The second part will focus on setting up an animated scene before you get into the weeds of timeline. Finally, the third section will discuss animating and more specifically ways to make your animations more visually pleasing.

I-Useful reminders, keybinds and basic functionalities

All the keybinds and features listed below are explained in the help section of the timeline UI but since I know none of you ever read it I am going to condense it down here with a bit of terminology as well.

A-Timeline UI



1. Frames per second.

2. Current time. Values can be typed in to move the cursor (vertical red line) to a given time.

3. Total duration of the animation.

4. Number of divisions in a given amount of time. The first number is a length of time, the second the number of divisions. Divisions show up as thin vertical lines on the timeline.

5. Time scale. Adjusts how fast the animation plays.

6. Single files can used to save/load existing animations.

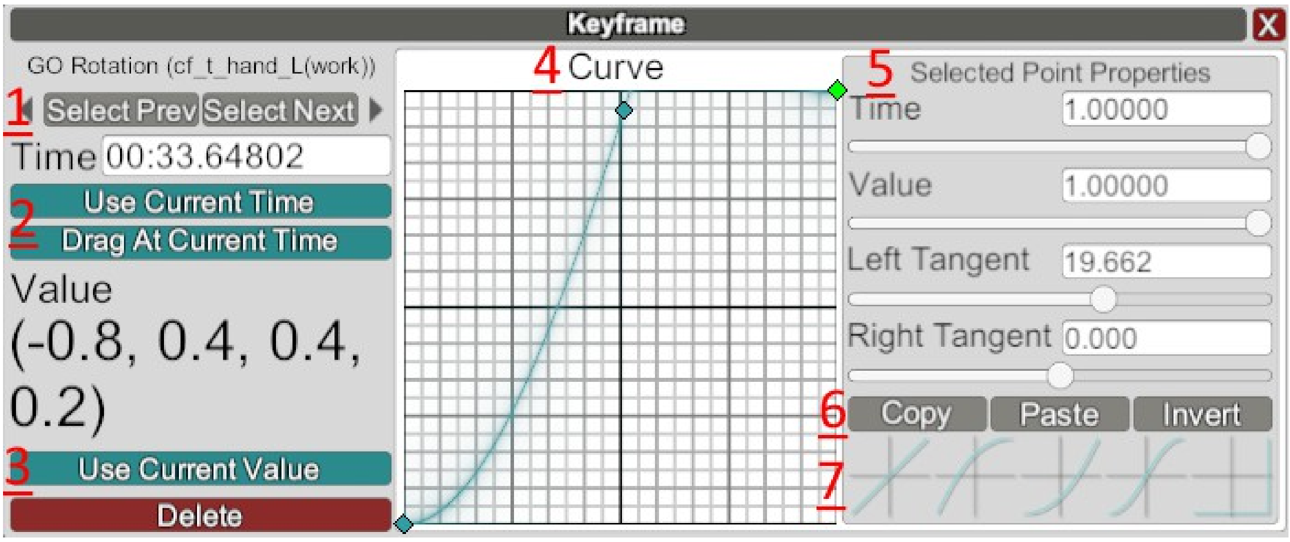
7. Toggle all interpolables. Self-explanatory.

8. List of interpolables. An interpolable is a value that can be changed through time.

9. Time scale. Left click on this bar to move the current time. Ctrl+MW to shrink/stretch the timeline.

10. Keyframes. Indicate values at a given time. Can be placed with middle mouse button (MMB). Can be removed with ctrl+MMB. Shift+MMB will snap the newly-created keyframe to the nearest division. Keyframes can be copied with ctrl+c and pasted at the current time with ctrl+v.

Clicking on a keyframe will turn it green and open this window:



1. Select last/next keyframe of the same interpolable

2. Use current time and Drag at current time are used to move the keyframe on the timeline. The keyframes will be moved to the current time indicated by the cursor. Drag at current time can be used when multiple keyframes are selected and will maintain the order or and time at which they appear relative to each other.

3. Use current value sets the value of the selected keyframes.

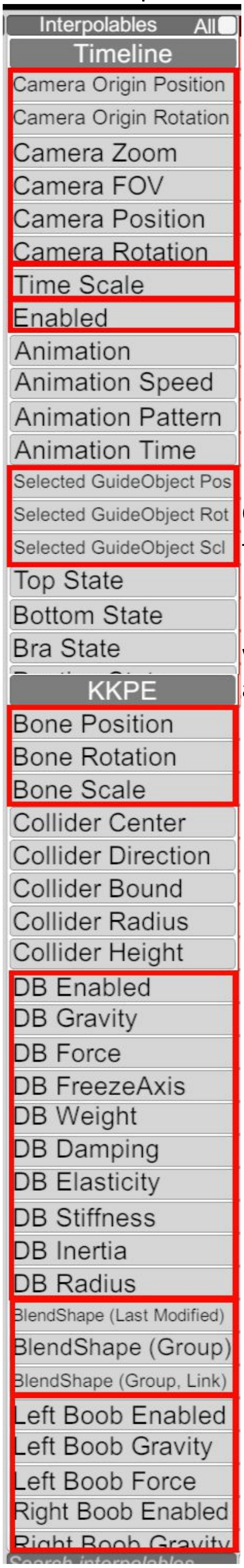
4. The curve indicates how the value changes to the next keyframes. The X-axis represents time where X=0 is the time of the currently selected keyframe and X=1 of the next keyframe. The Y-axis represents the values. Y=0 is the value of the selected keyframe and Y=1 of the next keyframe. The importance of the curve feature will be discussed further below.

5. Point properties of the curve. Points can be placed/remove on the curve and their tangents can be adjusted by these sliders of the mouse wheel.

6. Copy/paste/invert the current curve.

7. A set of predetermined curves. Clicking on each will replace the current curve.

Timeline has a wide array of interpolables and this list changes with what is currently selected in the workspace. Most are self-explanatory by their name alone, but here is a non-exhaustive list of the most important and commonly used interpolables in animations.



-Camera position/rotation/zoom/FOV: is used to move the camera during the animation. Is always present.

-Time Scale: used to adjust the animation's time scale, speeding up or slowing down the animation.

-Enabled: enables/disable a selected item. Useful to make objects appear/disappear.

-Selected Object Pos/Rot/Scale: the most commonly used interpolable. Controls the position, rotation and scale of the currently selected item in the workspace.

The KKPE section becomes available when a character is selected in the workspace and contains multiple useful features for intermediate and advanced users.

-Bone Position/Rotation/Scale: Controls the pos/rot/scale of the last selected bone in KKPE's advanced mode menu.

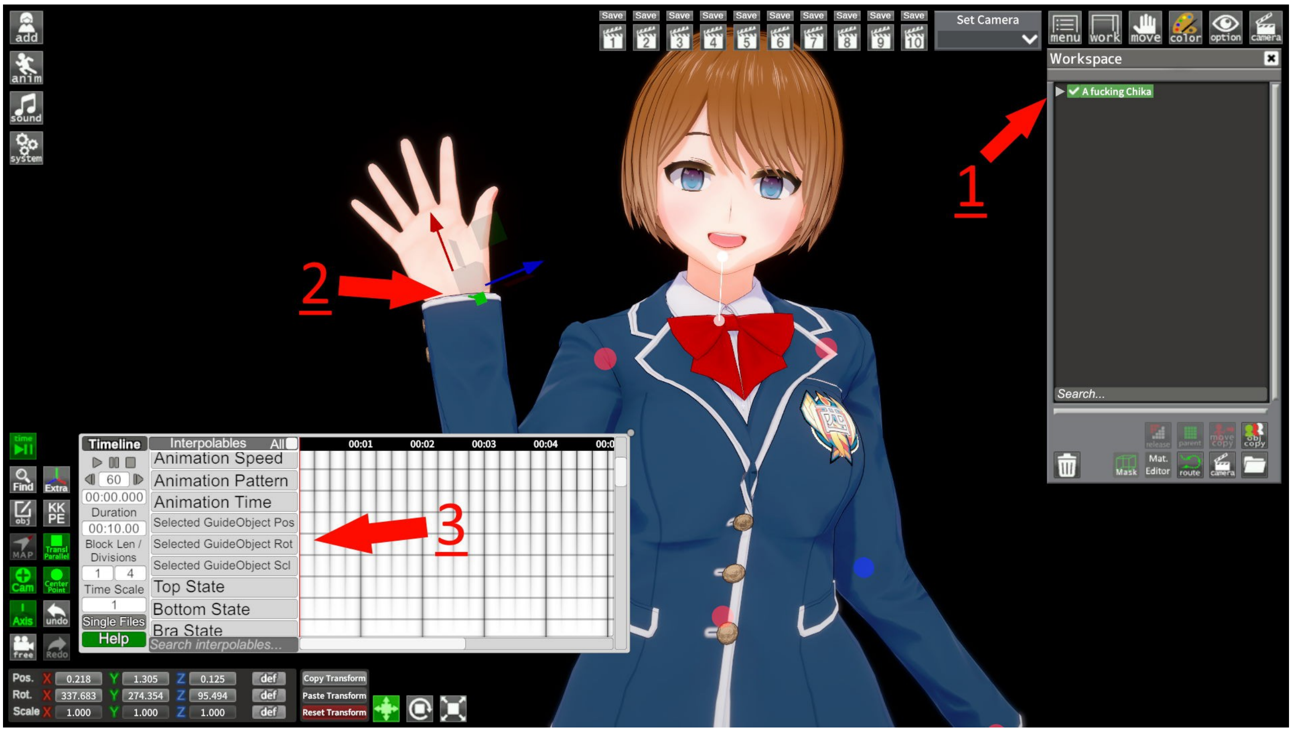
-Dynamic Bone (DB) controls: Adjusts the properties of the last selected Dynamic Bone.

-Blendshape controls: can be used to change the values of the last selected blendshapes. Last modified only affects one blendshape. Group affects all the blendhsape in one category (Eyes, Teeth, Tongue etc). Group Link affects the entire group and all the related blendshapes. Can be used to dynamically change a character's expression.

-Similar to DB controls but specifically used for the boobs and butt DBs.

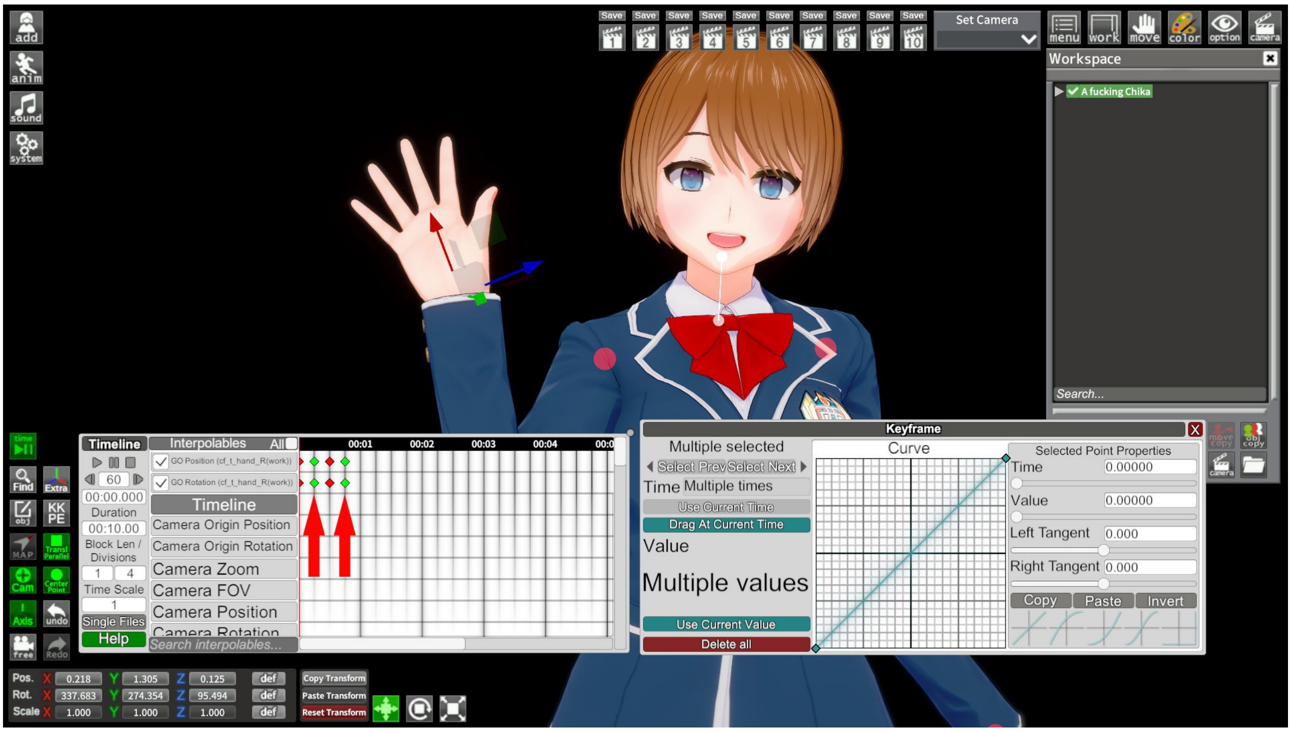
B. How to use

This section will be specifically dedicated to explaining how to get started with Timeline. A step-by-step description of how to make an entire animation will be given below, this part will focus on how to interact with Timeline. We will go over the basics with a simple animation of our character waving at the camera. This animation will only require une repetitive hand movement and is a good starter to get used with the mechanics of Timeline.



Start by posing your character as you normally would. FKIK is highly recommended for animations. Make sure to select it in the Workspace (1). In order to move our character's hand, we must first select it (2). Then, scroll down in Timeline to the selected pos/rot/scale interpolables and MMB in the empty timeline next to it (3). changes will not be immediately apparent, but you should see the list of interpolables move down by one position.

Scrolling back up to the top of the list will reveal your interpolables with one keyframe each. It is now time to place our keyframes for the animation. We want our character to move its hand back and forth with a fixed timing.
We will therefore place a series of keyframes. Shift+MMB will help to expedite the process. Do not forget to change the divisions. The default of 10 10 is not convenient for this application, I suggest a division of 1 4, one every 250ms.
Ctrl+MMB can be used to remove a keyframe. Make sure NOT to press del, as it will delete the selected character and all its keyframes.



With our keyframes in place, it is now time to change their values. Select every other keyframe as show above with ctrl+LMB. You can either click each one individually or drag over them. Doing so will open the keyframe window. Change the position and rotation of your character's hand and hit “Use Current Value” to replace the existing value with the new one.
You should now have a functioning, albeit basic, animation. You can extend it by selecting all the keyframes, copy them (ctrl+c), move your cursor over and paste (ctrl+v).
Currently our animation is very basic and looks stiff. The next two parts of this guide will go over how to make more complex and better-looking animations.

II-Setting up a Timeline Scene

The following sections will discuss how to set up your scene to get better results and to save time while animating. We will cover the entire process of making an animation from start to finish and in doing so we will go over various techniques and principles which you may find useful in your own works.
A well-made scene requires extensive preparation. This preparation will help you greatly in the long run to make your scenes more readable, easier to adjust and extend.

A-Nodes Constraints and parenting

Nodes Constraints (NC) is a tool which serves a similar function as the in-game parenting, but with a few significant differences. It is often used in animations for a number or reasons that will be discussed in the next section. In this section we will go over the main differences between NC and parenting as well as how to use this tool.

The NC menu is opened by default with ctrl+i. Here is the UI:

-1: toggles the position/rotation/scale constraint

-2: offset value between the parent and the child

-3: use the current offset between the parent and the child for the constraint.

-4: create a new constraint between the parent and the child. The child follows the parent with the parameters selected above.

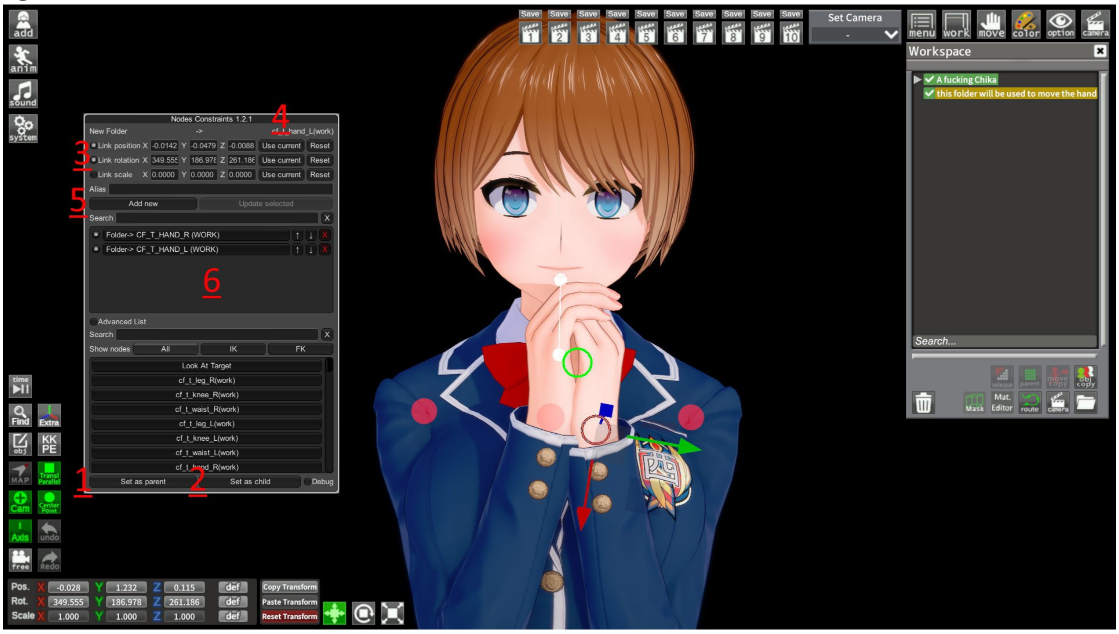
-5: update the selected constraint with the parameters selected above.

-6: set the selected item as parent/child. A green circle will indicate the parent and a red circle the child.

Why is NC important? A couple of key differences set it apart from parenting:

1. NC can affect bodyparts whereas parenting cannot.
2. NC can affect either position, rotation or scale or a combination of the above. Parenting affects all no matter what
3. In parenting the child can move independently within the reference frame of the parent. NC constraints are more rigid and do not allow for independent movement.

These properties mean NC is better suited to move bodyparts without the need for manual animating. This feature will come in handy later, but for now here is a quick step-by step description on how to use NC. As an example, let us attach our character's hands to a folder, allowing use to move both hands at once.



1. Select the folder and hit “Set as Parent”
2. Select one hand and hit “Set as Child”
3. Enable “Link Position” and “Link Rotation”. You can disable one of the two if you want the hands to react to the folder's position/rotation only.
4. Hit “Use Current” for both
5. hit “Add New”. The new constraint should show up in the window below
6. Repeat the process for the other hand.

Our character's hand should now be following the guide item. We have used a folder for this example but it can be anything: an object or even another bodypart.

B-Using folders for convenience

Perhaps the primary use of folders in animations is to move multiple bodyparts at once. Some, like the hips or shoulders are particularly suited for this: their fixed spacing means they often move in unison. Using a folder to control them has three advantages:

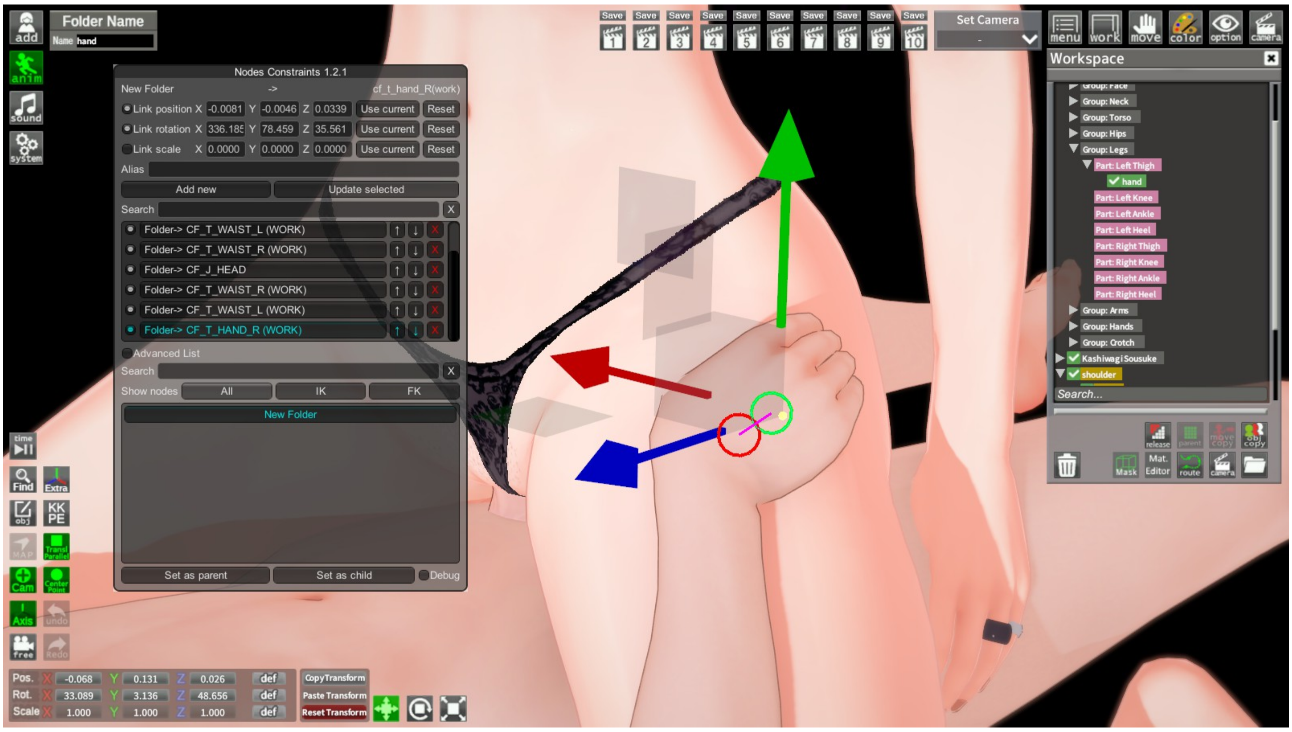
1. Fewer keyframes are needed and changing the TL values is made easier.
2. There is no risk of uneven/asymmetrical movement where it is not intended.
3. If properly done it is still possible to make more adjustments. In this example the child folder (named New Folder) is NC to the bodypart described above (hips of the 1st character, shoulders of the 2nd etc). The parent folder is moved with Timeline but the child remains unaffected. This child folder can thus be moved to accommodate smaller/taller characters without breaking the animation.

Additionnally you can see in the example shown above folders named “head1” and “head2”. These folders are used to rotate the character's heads. Head 2 is NC (rotation only) to the head. When the folder is rotated, the head follows. When head 1 and 2 rotate, their rotations are combined and transferred to the character's head. This method can be used for nearly any bodypart and allows for more complicated movement (circular motion, superimposed animations etc).

We will now create a new animation from scratch as an example. First things first, I highly recommend you look up reference images/videos for what you are trying to make. Doing so will help you decompose the movement you are trying to recreate. Try to figure out what moves and what can and should be controlled by folders. In our case, our characters' hips and shoulders can be moved with the help of a folder. The same goes for the heads.
Generally it is possible to use folders for the shoulders, hips and legs.
We will start by placing our folders and NC them to our characters. Focus on one of the shoulder/hip bone with ctrl+f and place a folder. Move the folder to the middle point between the two joints. It is generally recommended to place the folder between the two points as it prevents weird behavior when rotating. The name of our folders used here does not have any effect on their functionality, feel free to name them as you wish.

It is also possible to use folders and NC to attach a bodypart to another one. Let's say in that case we want to attach the male's hand to her leg. Parent a folder (an obect would do as well) to her leg and NC his hand to that folder as depicted below. Doing so early on makes it so you don't have to manually animate all these parts. Note however that this method does not work with dynamic bones such as the ass or breasts.
Now that our scene has all the required setup it is time to animate it. The next section will discuss how to make your animation appear smoother, natural or stylized.

Here you can get a preview of the finished animation. I did not animate the face as this is not the focus of this guide: https://files.catbox.moe/fdpr76.webm
The scene can be found here: https://files.catbox.moe/z06oo7.png

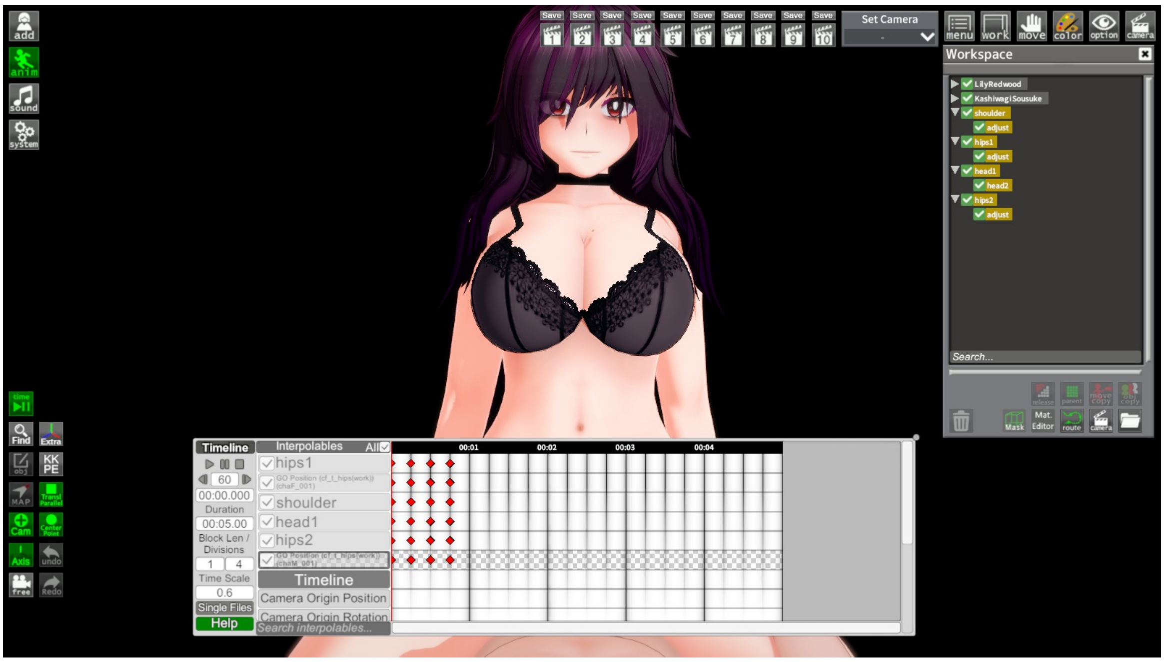


III-Animating quickly and efficiently

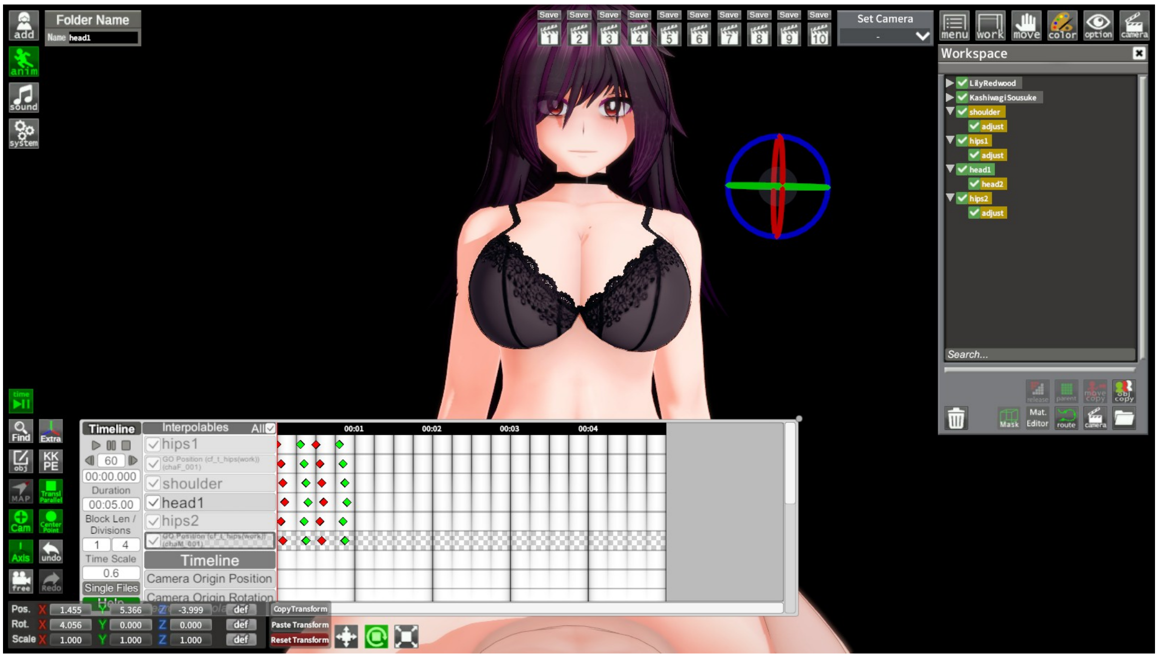
A-Timing and delay

One of the easiest way to make your animations appear more natural is to work on the timing of your keyframes. This is because be it in the real world or animations not everything moves in unison all the time. It is often easier to think of what initiates a movement and what moves as a reaction afterwards. This is another reason why studying other animations before you begin can be useful as it can help you figure this out as well.

Start by placing your keyframes using the divisions for simplicity. They can always be moved later with “Drag at Current Time”. I suggest you work with a small set of keyframes at first and copy/paste your short loop once you are satisfied with the result.



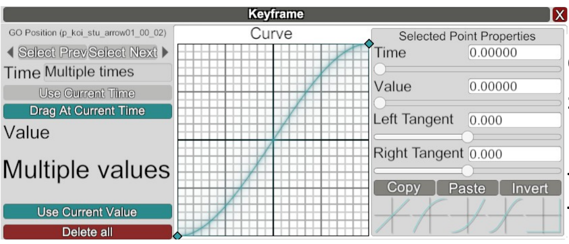
You can do two things here. You can either add delay between your various interpolables or you can move a whole set of keyframes.
The former will make the animation appear smoother and more natural, the latter will speed up or slow down a section of your animation. In that case we want the down stroke of the movement to be faster so we will give more time to the up stroke.
In this animation the hip is what initiates the movement, followed by the waist, the shoulders and finally the head. The other character's hips also reacts. We ill thus add delay to each interpolable in that order.



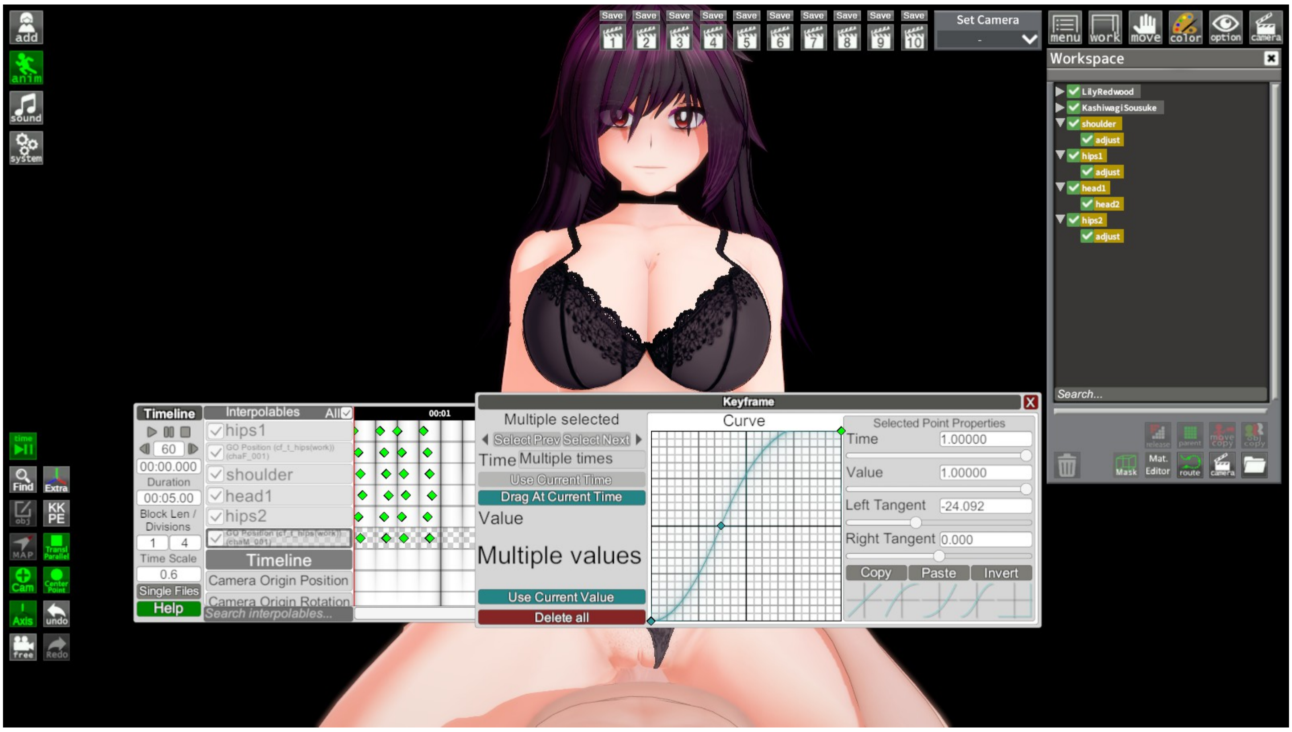
If your animations still appears unnatural after this step, try working on the order or the length of that delay. Excessive delay can make things appear strange. The longer the delay, the more pronounced the effect is, so it is best to use it sparingly. In that specifc case, I added a 50ms delay between the hips and the waist, 25ms between the waist and the shoulders and another 25ms between the shoulders and the head. However you can only get so far with delay alone and working on your keyframes' curves can help immensely.

B. Curve editing

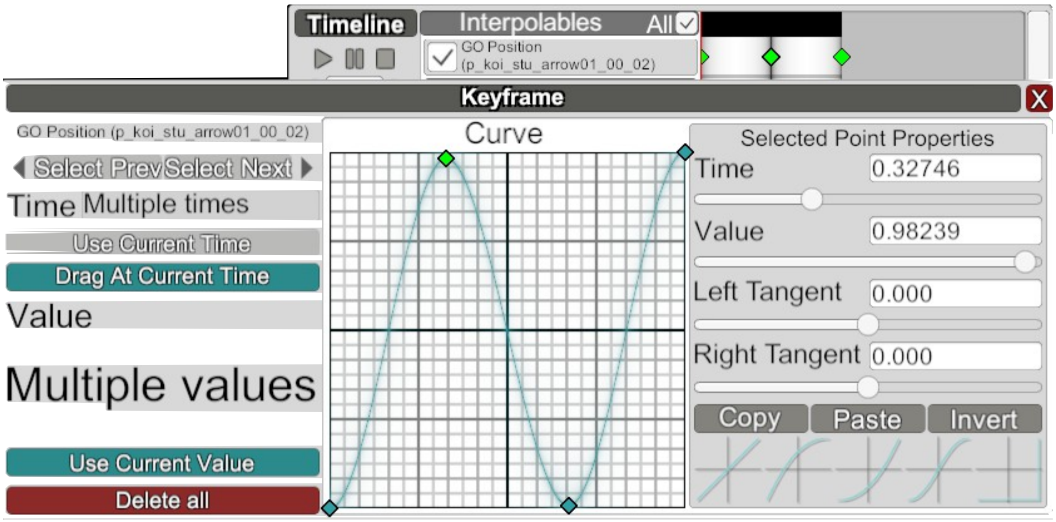
Perhaps the most important part of any animation is the curve. Understanding this feature and what it does will immediately lead to better results. In short, the Curve is here to define the speed of the change between the current keyframe and the next. It does so by having two points, a start and an end point, and a curve linking the two. The X-axis is time, and the Y-axis is the value of the interpolable. A different curve shape implies a different acceleration. A flat curve describes a slow movement while a steep curve describes a rapid change.



Most beginners use the default smoothed curve. This curve simulates acceleration and decceleration and is superior in most instances to the default flat curve. However, it is still quite robotic and stiff as very few things in real life follow that sort of smooth transition. Things in real life start and stop abruptly, bounce and get dragged on by inertia. It can thus help to make the curve asymmetrical by adding a third point. Inertia in particular can help greatly with making your animations appear more expressive.

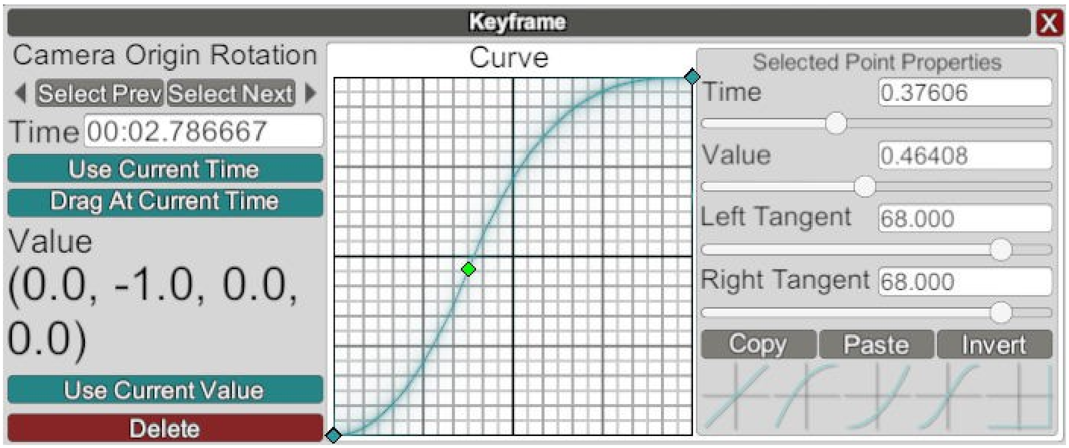


The easiest way to replicate inertia in your animation is to adjust the tangents of your 2nd and 3rd point (end point) so that the curve goes outside the graph window. Indeed the curve is not limited to values between 0 and 1. Even though you cannot see it, the curve Y-position exceeds 1, causing the value to overshoot. This also demonstrates that the curve does not always have to go up. This can be used to create the same effect as the default smooth curve while using fewer keyframes as shown below.

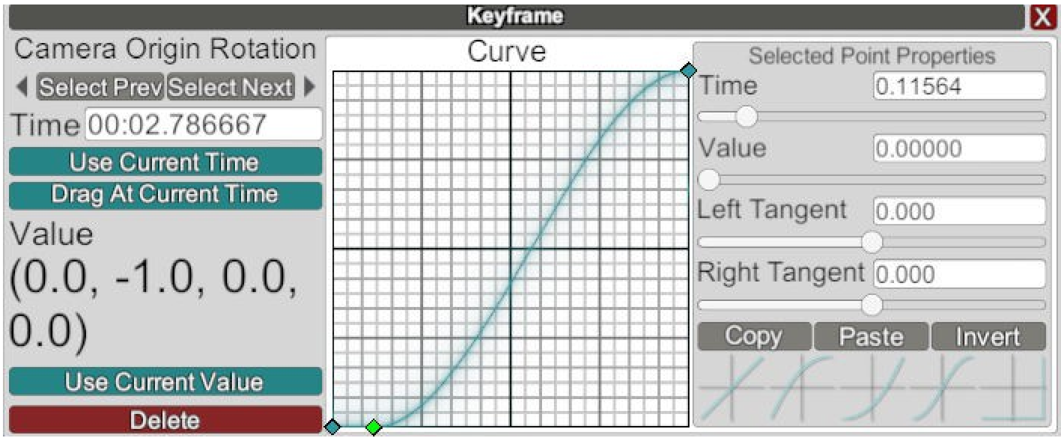


Finally here is a non-exhaustive list of curves you may use in your animations and what they do. Choose which one to use wisely.

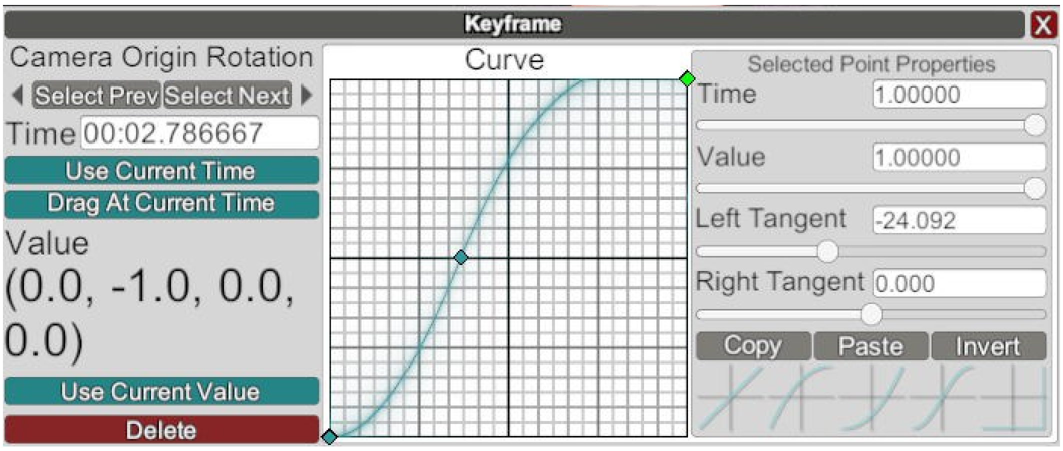
Offset smoothed curve: works best for slow movements. Adds a bit of variety compared to the default smooth. Add a point to the lft or right of the center and adjust the tangnt with your mousewheel until it looks smooth.



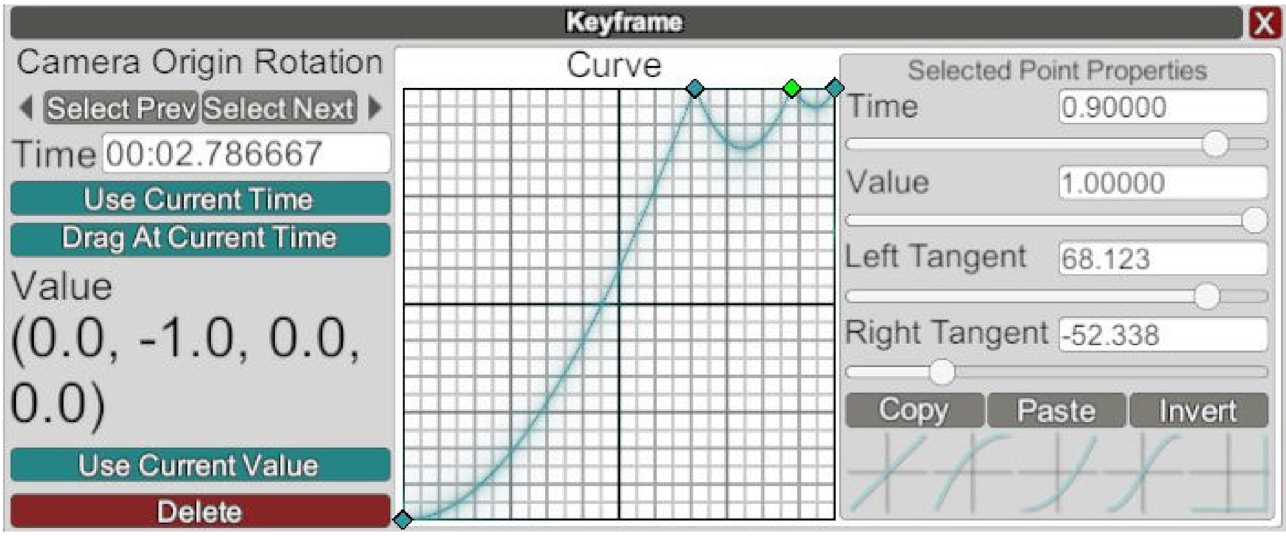
Offset smoothed with a pause: same as above but with a small pause at the beginning or the end. Gives a more cartooney feel to your animations. Keep the pause short.



Smoothed with overshooting curve: works best to mimic intertia. Same as the offset smoothed but the left tangent on the end point is adjusted. The lower the value, the more pronounced the effect.



The bounce: used to recreate the bounce of an object against a hard surface. Works best ith falling objects. Add points near the end of the curve at Y=1 and adjust their tangents. You may add more or fewer bounces depending on the physical properties you want to recreate. Make sure each cosnecutive bounce is faster and smaller.



Finally you can combine the various curves shown above for your animations.